

GIOIA DEL COLLE, Italy

WMO: 163120

Lat: 40.766N

Lon: 16.935E

Elev: 1154

StdP: 14.09

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.5	31.8	21.3	16.6	37.0	24.7	19.6	37.5	37.0	50.7	31.6	48.6	7.2	320	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	94.8	70.3	91.5	69.2	88.7	68.7	74.6	84.2	73.0	82.6	71.6	81.2	11.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.0	123.6	77.3	70.2	116.0	76.3	68.5	109.2	75.1	38.9	84.4	37.4	83.1	36.2	81.3	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
30.7	26.1	22.1	DB	23.9	100.9	2.9	3.7	21.8	103.6	20.2	105.7	18.6	107.8	16.5	110.6
			WB	23.0	78.0	2.5	2.0	21.2	79.5	19.7	80.7	18.3	81.8	16.5	83.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.8	43.5	44.1	48.6	54.4	62.7	71.4	76.3	76.3	68.0	60.6	52.7	45.7
	DBStd	12.86	5.13	5.36	5.18	4.82	5.34	5.85	4.59	4.37	5.33	5.30	5.73	5.37
	HDD50	678	207	173	88	16	0	0	0	0	0	2	38	154
	HDD65	3349	666	585	509	319	108	13	0	0	28	154	368	599
	CDD50	3882	6	8	44	147	394	643	815	814	541	330	120	20
	CDD65	1079	0	0	0	1	38	206	350	349	119	16	0	0
	CDH74	10239	0	0	1	19	384	2003	3642	3371	763	56	0	0
	CDH80	4226	0	0	0	0	80	786	1649	1496	213	2	0	0
(14) Wind	WSAvg	9.3	9.7	10.6	10.8	10.1	8.8	8.4	8.9	8.1	8.4	8.4	9.9	9.5
(15) Precipitation	PrecAvg	26.5	2.6	2.8	2.8	1.7	1.9	1.6	1.1	1.7	2.0	2.8	2.7	2.8
	PrecMax	38.9	5.7	8.5	6.8	4.2	6.0	7.4	4.4	5.7	5.9	6.9	6.3	4.7
	PrecMin	12.7	0.4	0.4	0.4	0.2	0.2	0.1	0.0	0.0	0.0	0.0	0.6	1.2
	PrecStd	8.6	1.6	1.9	1.7	1.1	1.5	1.7	1.2	1.7	1.7	1.9	1.9	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.9	61.7	68.4	76.9	86.4	95.3	100.1	99.0	91.3	79.0	69.7	61.1
		MCWB	52.3	51.4	54.1	59.7	66.3	70.7	70.2	70.4	67.3	64.6	59.7	55.0
	2%	DB	55.8	57.7	64.4	70.2	82.1	91.4	95.0	94.7	85.7	75.2	66.1	58.1
		MCWB	50.8	49.8	53.3	57.2	63.9	69.0	70.4	70.2	66.9	63.6	59.6	54.1
	5%	DB	53.8	55.4	60.9	67.5	78.5	87.9	91.5	91.3	82.0	72.0	63.0	56.5
		MCWB	49.9	49.4	52.0	56.0	62.3	68.1	68.9	69.8	66.4	62.6	58.5	53.1
	10%	DB	51.9	53.3	57.5	64.2	75.0	84.4	88.2	87.9	78.4	69.6	61.0	54.2
		MCWB	48.6	48.7	50.7	54.9	61.4	66.9	68.3	69.0	65.5	62.5	57.4	51.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.3	54.6	57.8	62.4	68.9	75.2	76.7	77.2	73.9	70.2	63.9	57.8
		MCDB	57.7	57.9	63.8	70.3	80.4	87.1	88.5	88.9	78.8	74.1	65.8	59.8
	2%	WB	52.9	52.7	55.3	59.6	66.6	72.4	74.3	74.8	71.7	67.5	61.6	55.6
		MCDB	54.8	55.7	60.5	66.7	77.0	85.0	84.9	83.8	77.5	70.5	63.7	57.4
	5%	WB	50.9	51.0	53.6	57.6	64.8	70.7	72.6	73.3	70.0	65.7	59.7	53.8
		MCDB	53.2	54.0	58.1	63.9	74.2	82.2	83.3	81.9	76.0	69.0	62.2	55.8
	10%	WB	49.1	49.4	52.2	56.1	63.1	68.7	71.1	71.8	68.3	63.9	58.1	51.9
		MCDB	51.5	52.3	56.5	61.6	71.5	79.7	82.3	80.6	73.9	67.6	60.6	54.2
(35) Mean Daily Temperature Range	MDBR		12.5	13.7	15.2	16.9	19.3	20.6	21.2	21.1	17.5	15.5	12.7	12.0
	5% DB	MCDBR	15.3	17.4	21.3	22.9	25.6	25.9	26.8	26.6	22.6	19.4	15.1	13.1
		MCWBR	11.2	11.5	12.6	12.4	11.2	10.0	10.0	10.8	9.8	10.9	10.5	10.1
	5% WB	MCDBR	12.4	13.9	17.1	18.9	21.8	22.6	22.2	21.6	18.4	15.6	11.9	10.6
		MCWBR	10.3	10.5	11.3	11.7	11.4	10.6	10.5	10.7	10.2	10.3	9.5	9.1
(40) Clear-Sky Solar Irradiance	taub		0.330	0.352	0.389	0.437	0.432	0.419	0.414	0.422	0.414	0.383	0.355	0.333
	taud		2.461	2.377	2.275	2.138	2.216	2.293	2.313	2.296	2.312	2.386	2.418	2.453
	Ebn at Noon		261	271	272	265	269	272	272	267	261	256	248	250
	Edh at Noon		25	31	38	47	45	41	40	40	37	30	26	23
(44) All-Sky Solar Radiation	RadAvg		587	827	1238	1623	1983	2238	2291	2014	1471	1029	667	524
	RadStd		60	74	125	126	118	116	97	117	85	82	55	62

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-166	N/A	N/A
(47)	Regional (11 neighbors)	+0.63	N/A	N/A	N/A	N/A	N/A	N/A	-121	+166	N/A

Nomenclature: See separate page